

Solar System Research: Evidence Desk

GROW • Science • 40 minutes

I can research one Solar System focus using suitable sources, accurate notes and a clear source record.

Prepare before pupils arrive

Setup: Keep the nine dated local source cards inside the lesson and print one full set per group. Provide a paper note grid and an optional local typing page.

Safety: No live web or personal account is required. Follow school privacy and device-handling routines for the optional local route.

Equipment: The nine local dated source cards in this lesson, a paper or local-page note grid and pencil or school device.

Safety: No website, account or personal information is needed. Use the local lesson page or printed cards and follow school device routines.

Fallback: Use the same nine dated source cards printed from the lesson. Write or scribe the notes on paper.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Research and present findings about the Solar System using ICT.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	Last lesson: why were three views stronger than one? Last week: state one model limitation from the shape lab.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 5 min	source — where information came from credible — reasonable to trust paraphrase — same meaning, your words Turn a broad topic into one answerable question, such as: How long does Mars take to orbit the Sun? Check whether a source names a science organisation or author, answers the question and gives information you can compare with another source.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Read one short source statement, look away, then write the key meaning in fewer words. Compare the note with the source: keep number, unit and relationship accurate; repair any changed meaning.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each research item to keep, check or reject, then freeze the decisions you can justify. Read one source-and-note card.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Choose one local Solar System question and prepare a three-fact research record for the next lesson. Record: One chosen question, three accurate own-word facts, at least two source codes, one agreement check and one card to check or leave out.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Explain why one chosen note is ready for a presentation and one other item is not. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: 'Mars takes about 687 Earth days to orbit the Sun' paraphrased, with the checked source recorded.

Correct. It answers the question, keeps the scientific meaning and records a source another person can find.

Misconception repair

Repaired. Select the relevant meaning, paraphrase without changing the Science and leave a source trail.

Guided checkpoint

Correct. It answers the question, keeps the scientific meaning and records a source another person can find. Look for relevance, accurate wording and a record that lets another person check the information.

Named science source

Keep for the record. Keep: the note answers the question, keeps the meaning and has a source code. This note passes the checks and another reader can find its source card.

Old page, changing claim

Check before use. Check: current mission information needs a visible date or a second up-to-date source. The source may be useful, but a fact about a current mission needs a date check.

Unnamed dramatic post

Reject for this question. Reject: the dramatic claim has no evidence and no named source. A striking claim without evidence or a named source is not ready to use.

Accurate but copied

Check before use. Check: keep the source open, paraphrase accurately and add the source record before use. The information may be useful, but the research note is not ready as written.

Reliable but off-topic

Reject for this question. Reject for this question: a trusted source still needs to answer the chosen question. Ask whether the item helps answer the exact research question.

Cross-checked fact

Keep for the record. Keep: agreement between two sources and clear source codes strengthen the note. This is the strongest ready-to-use record in the set.

Model limitation

Checking a source helps, but it does not guarantee that every statement is right. Facts may be shortened, updated or given with limits.

Independent evidence

One chosen question, three accurate own-word facts, at least two source codes, one agreement check and one card to check or leave out.

Supported: Choose one question and use its three local cards to complete three short source-note rows. Use highlighted key words and: The card says ___. This means ___. Source code ___.

Standard: Choose one question, compare its three local cards and prepare three facts in your own words. Record source code, key meaning, own-word note and use/check/leave-out choice in separate boxes.

Stretch: Research a question with a comparison or explanation, cross-check claims and note any uncertainty or disagreement. Evaluate why each source is suitable and preserve exact values, units and qualifications.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_G_W10A_Solar_System_Research_Explore.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Source references embedded in the uploaded lesson:

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{
  "id": "saturn-rings",
  "question": "What are Saturn's rings made of?",
  "cards": [
    {
      "sourceCode": "SAT-A",
      "publisher": "NASA Science · local adapted fact card",
      "date": "Pack checked 27 August 2026",
      "text": "Saturn's rings contain billions of separate pieces of ice and rock. The pieces range from tiny grains to much larger chunks."
    },
    {
      "sourceCode": "SAT-B",
      "publisher": "European Space Agency Education · local adapted fact card",
      "date": "Pack checked 27 August 2026",
      "text": "Water ice makes up much of Saturn's bright rings. Dust and rocky material are mixed with the ice."
    }
  ]
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{"sourceCode": "SAT-C", "publisher": "Royal Museums Greenwich Education · local adapted fact card", "date": "Pack checked 27 August 2026", "text": "Saturn's rings are many narrow ringlets, not one solid sheet. Icy pieces reflect sunlight, helping the rings look bright."}}
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{"id": "mars-red", "question": "Why does Mars look red?", "cards": [{"sourceCode": "MAR-A", "publisher": "NASA Science · local adapted fact card", "date": "Pack checked 27 August 2026", "text": "Iron minerals in Martian rock and soil reacted with oxygen over time. The iron oxide produced is like rust and gives much of the surface a red colour."}, {"sourceCode": "MAR-B", "publisher": "European Space Agency Education · local adapted fact card", "date": "Pack checked 27 August 2026", "text": "Fine dust rich in iron oxides covers much of Mars. Winds can lift this red dust into the thin atmosphere."}, {"sourceCode": "MAR-C", "publisher": "Royal Museums Greenwich Education · local adapted fact card", "date": "Pack checked 27 August 2026", "text": "From Earth, Mars has a red-orange appearance. Iron oxide in its surface dust is the main reason for this colour."}]}
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{"id": "earth-mars-years", "question": "How long do Earth and Mars take to orbit the Sun?", "cards": [{"sourceCode": "ORB-A", "publisher": "NASA Science · local adapted fact card", "date": "Pack checked 27 August 2026", "text": "Earth takes about 365.25 Earth days to orbit the Sun. Mars takes about 687 Earth days."}, {"sourceCode": "ORB-B", "publisher": "European Space Agency Education · local adapted fact card", "date": "Pack checked 27 August 2026", "text": "Mars is farther from the Sun than Earth and its orbit takes longer. One Martian year is nearly two Earth years."}, {"sourceCode": "ORB-C", "publisher": "Royal Museums Greenwich Education · local adapted fact card", "date": "Pack checked 27 August 2026", "text": "Earth is the third planet and Mars is the fourth. A Mars year lasts about 687 Earth days, compared with about 365 days for Earth."}]}
```

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.